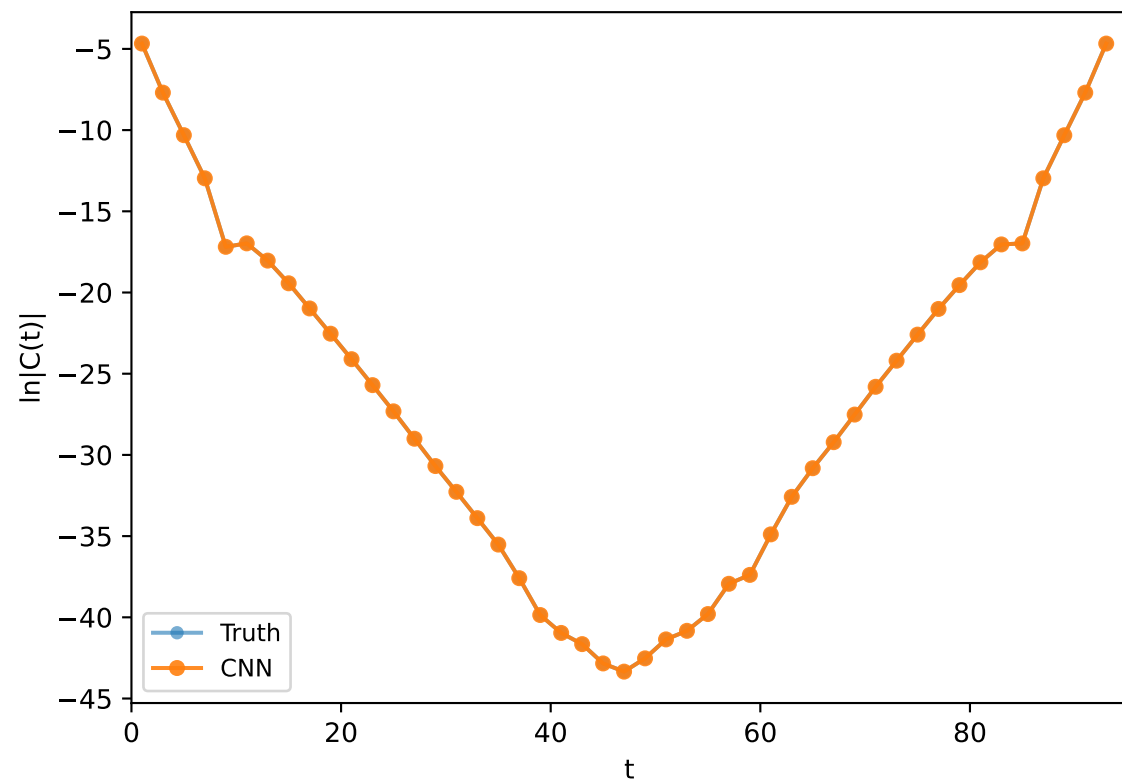


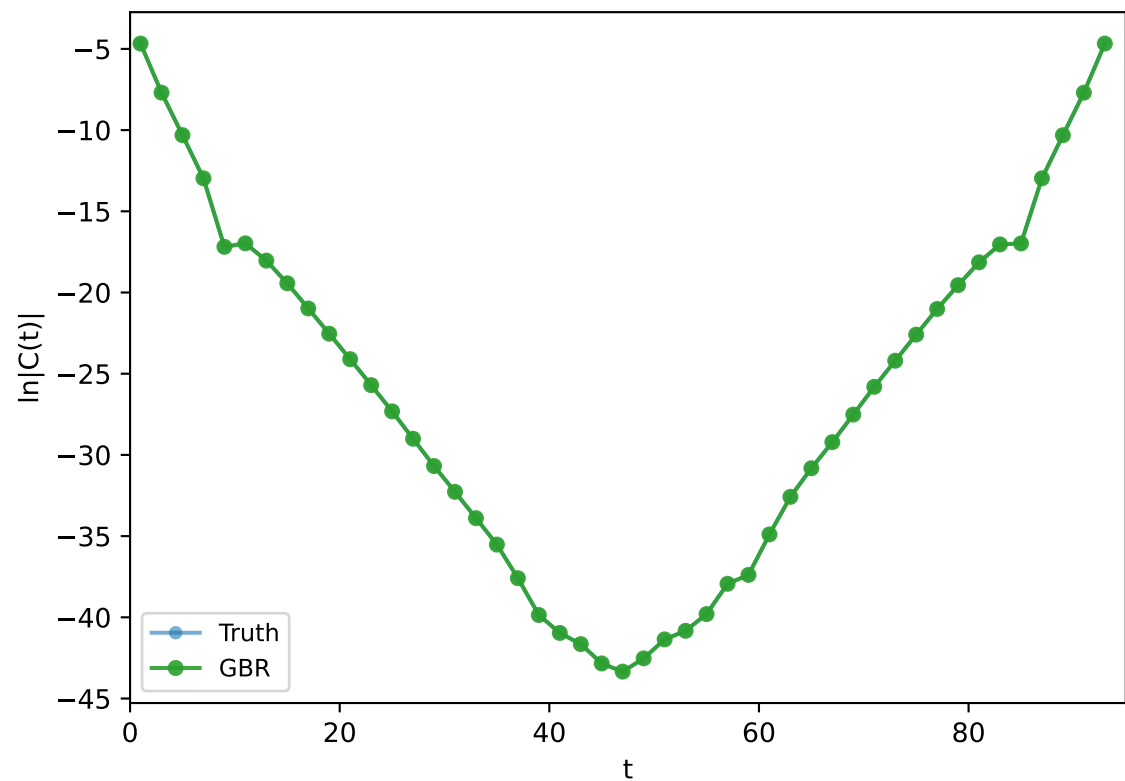
Two_pt_ML_OTSP_D_gold_to_nongold_Experiment

Log Correlator Comparison

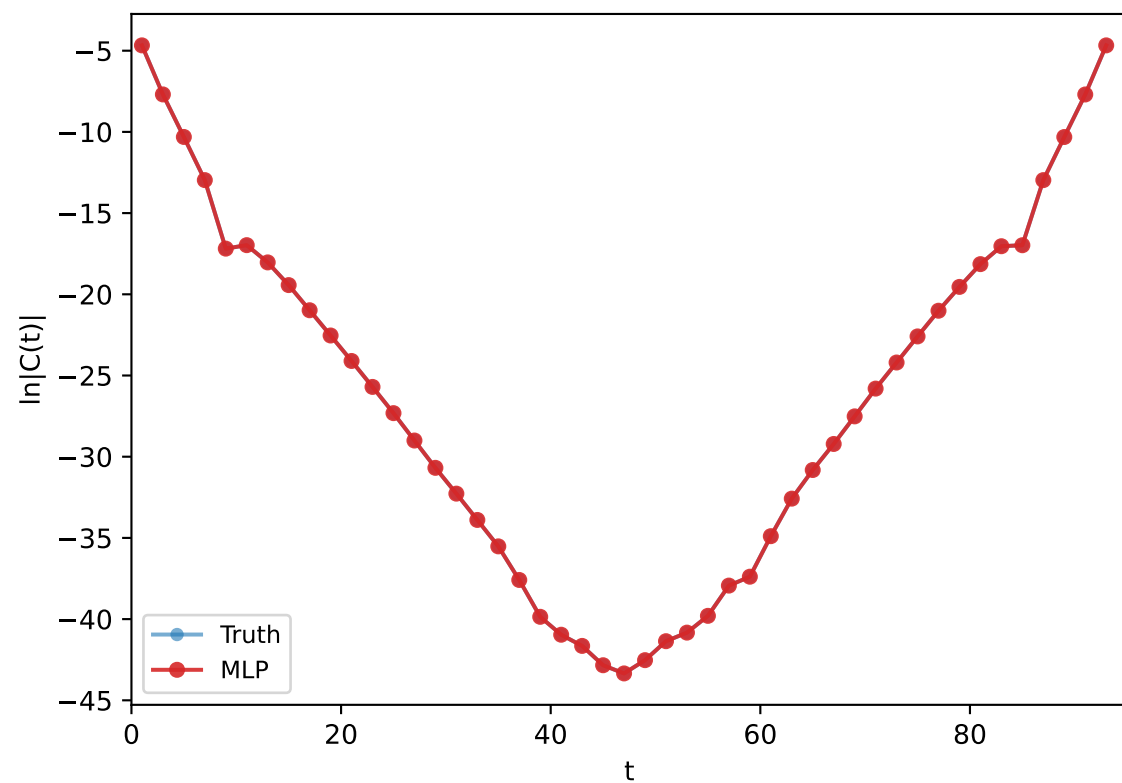
CNN vs Truth



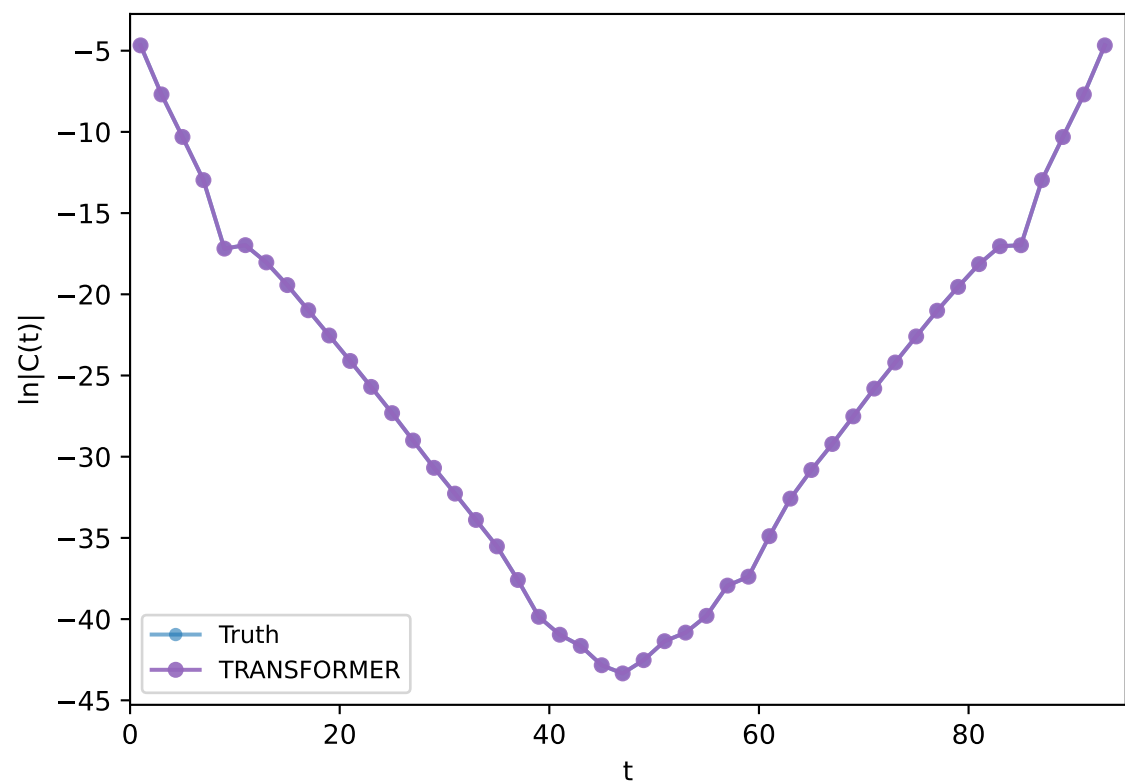
GBR vs Truth



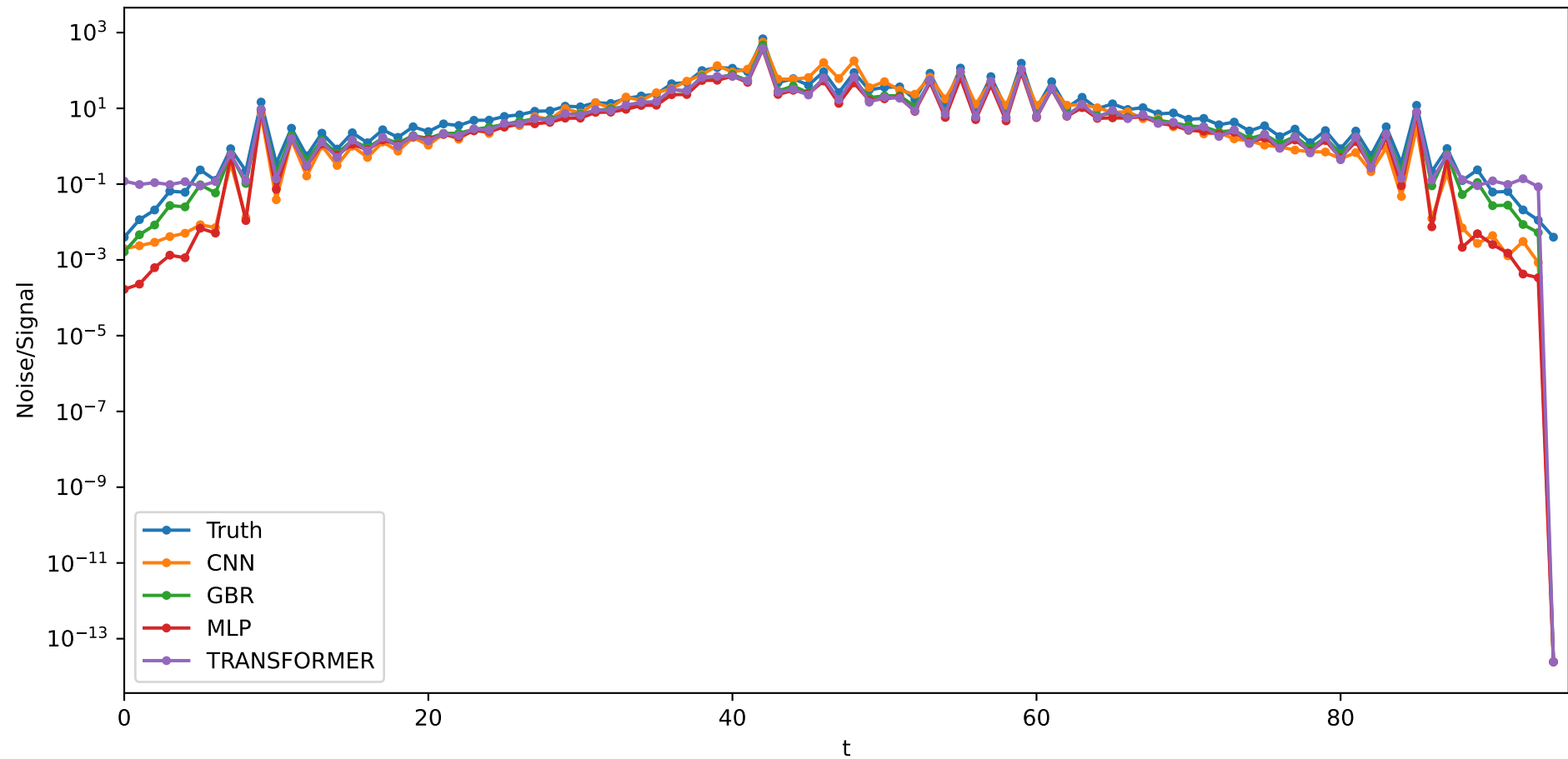
MLP vs Truth



TRANSFORMER vs Truth



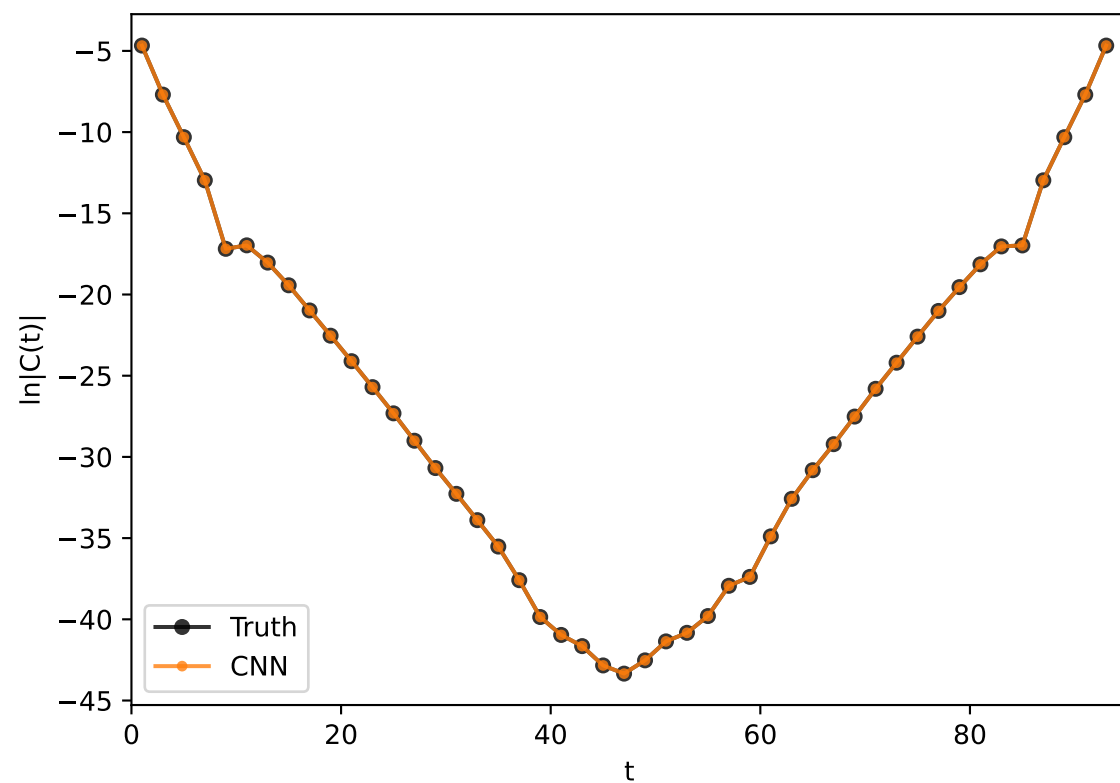
Two_pt_ML_OTSP_D_gold_to_nongold_Experiment
Noise-to-Signal Ratio



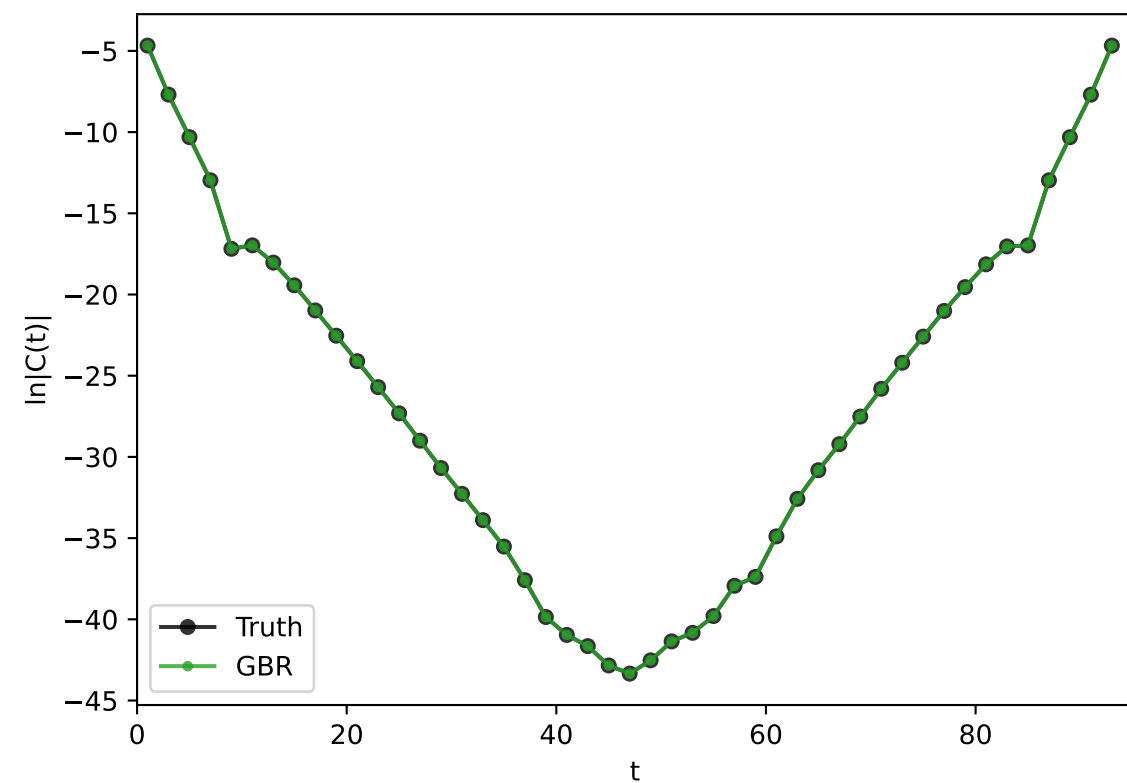
Two_pt_ML_OTSP_D_gold_to_nongold_Experiment

Bias Correction Effect

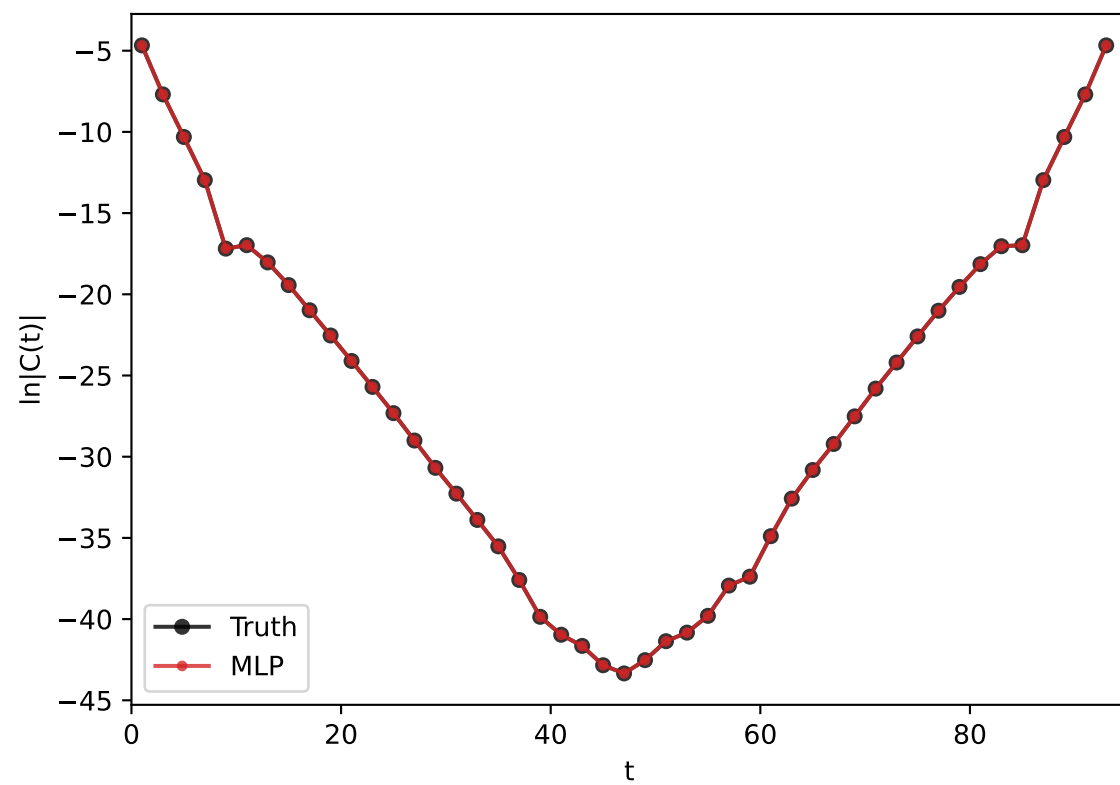
CNN: BC



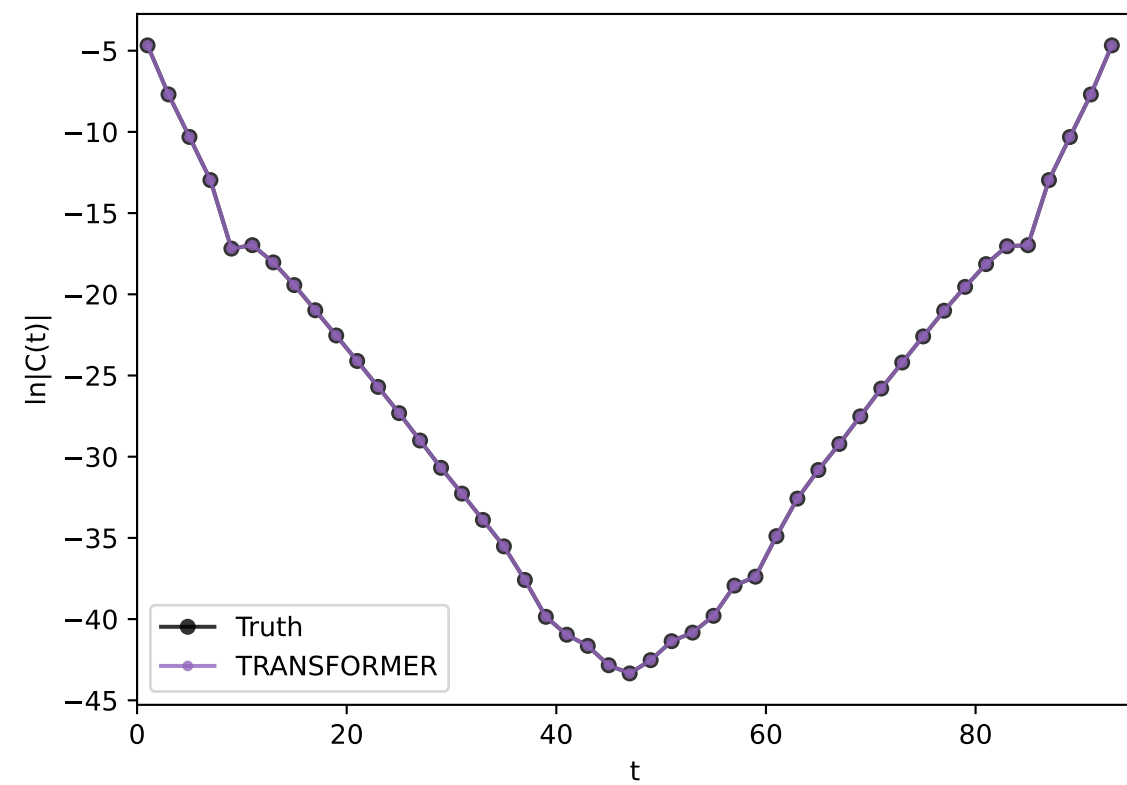
GBR: BC



MLP: BC

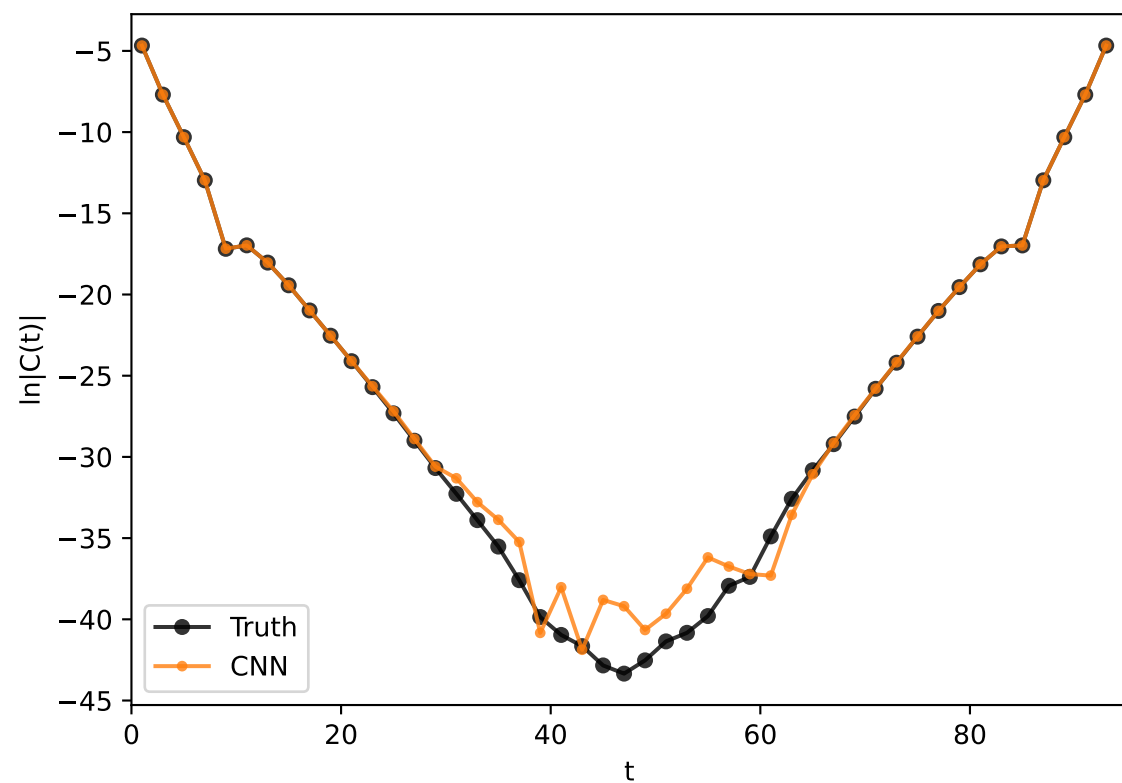


TRANSFORMER: BC

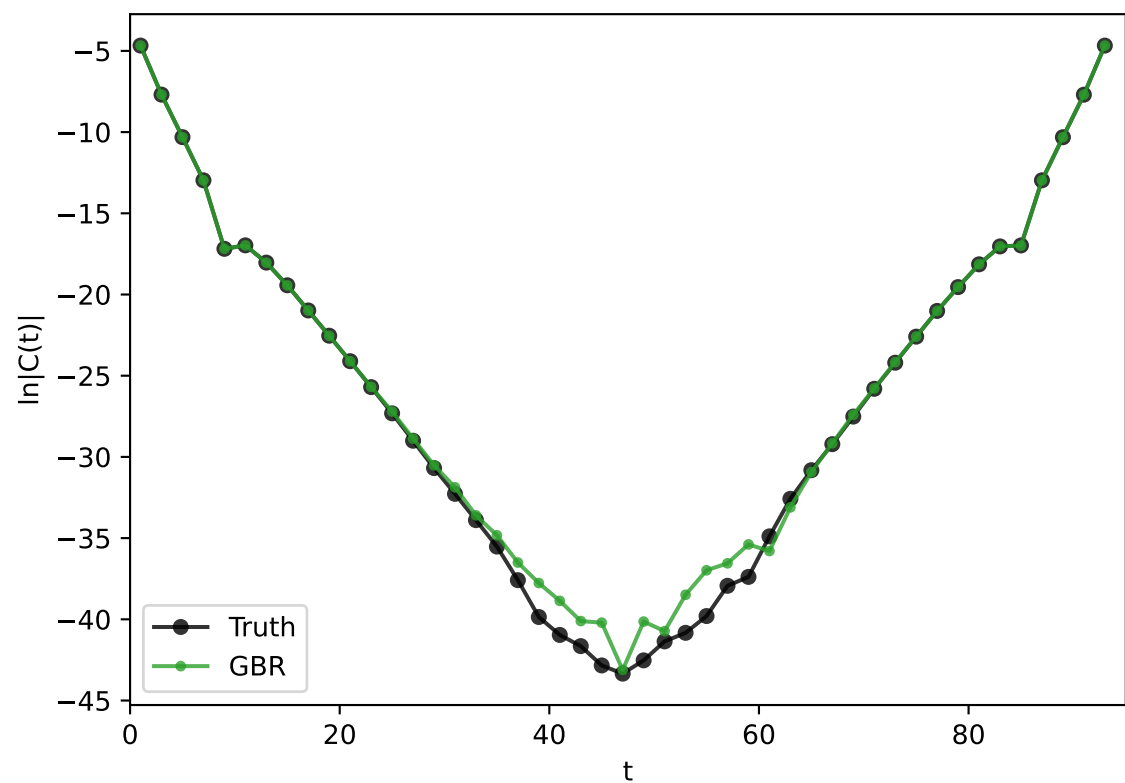


Two_pt_ML_OTSP_D_gold_to_nongold_Experiment
Ratio Method

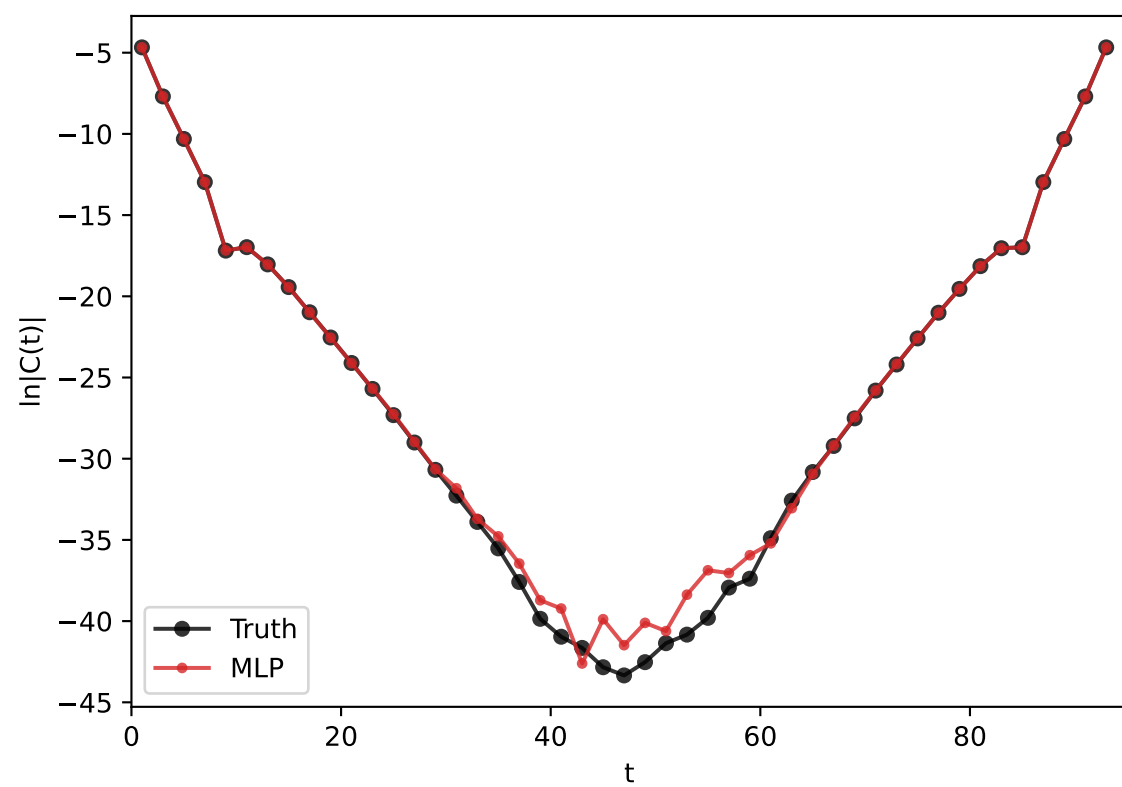
CNN: RM



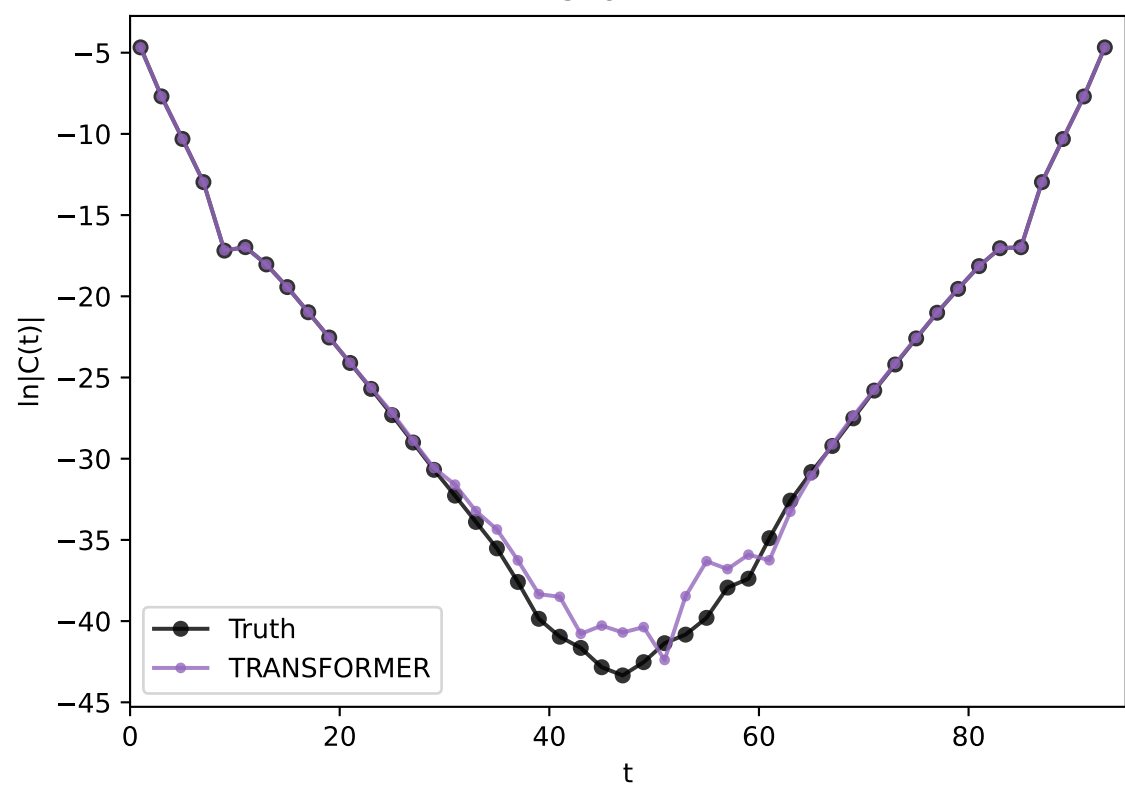
GBR: RM



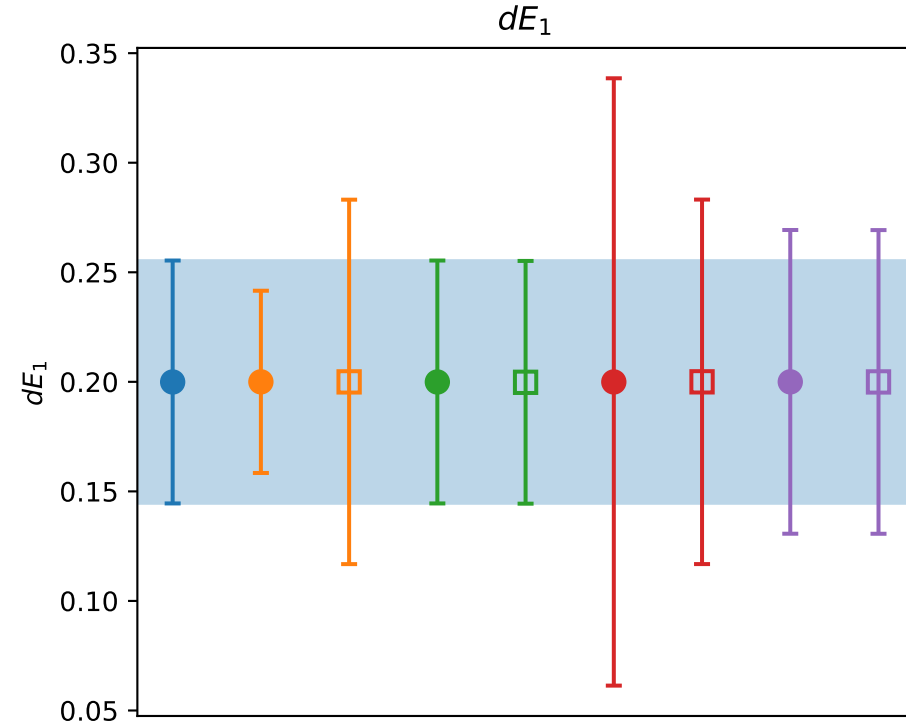
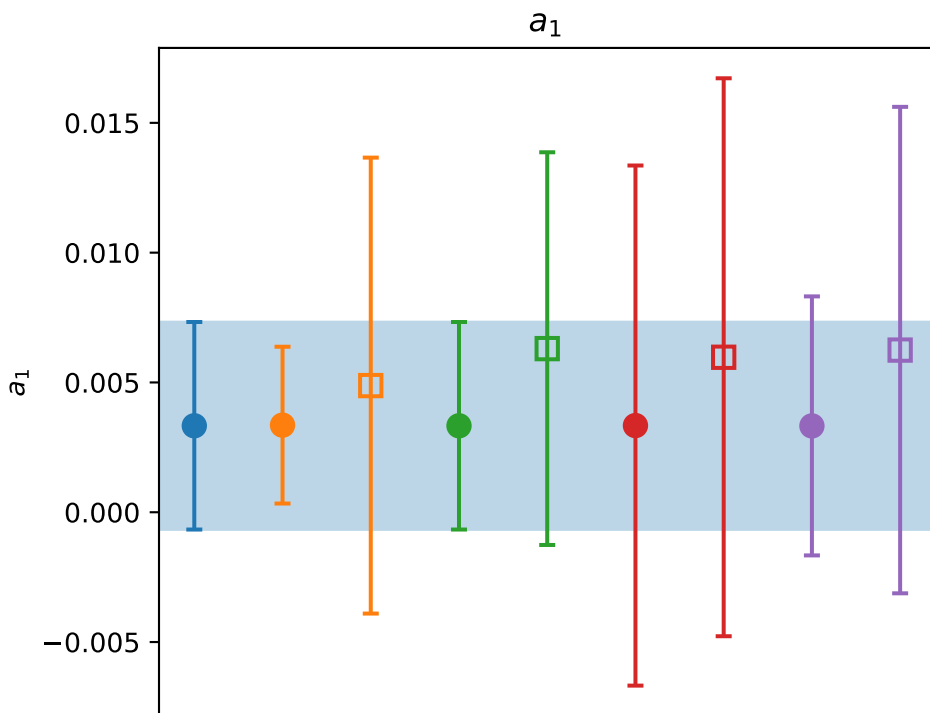
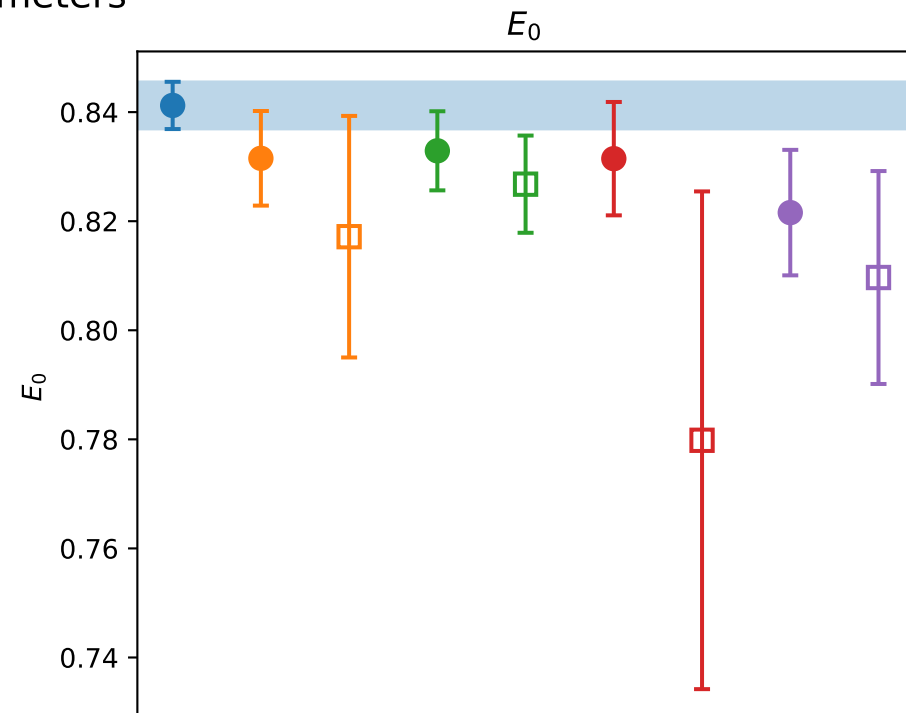
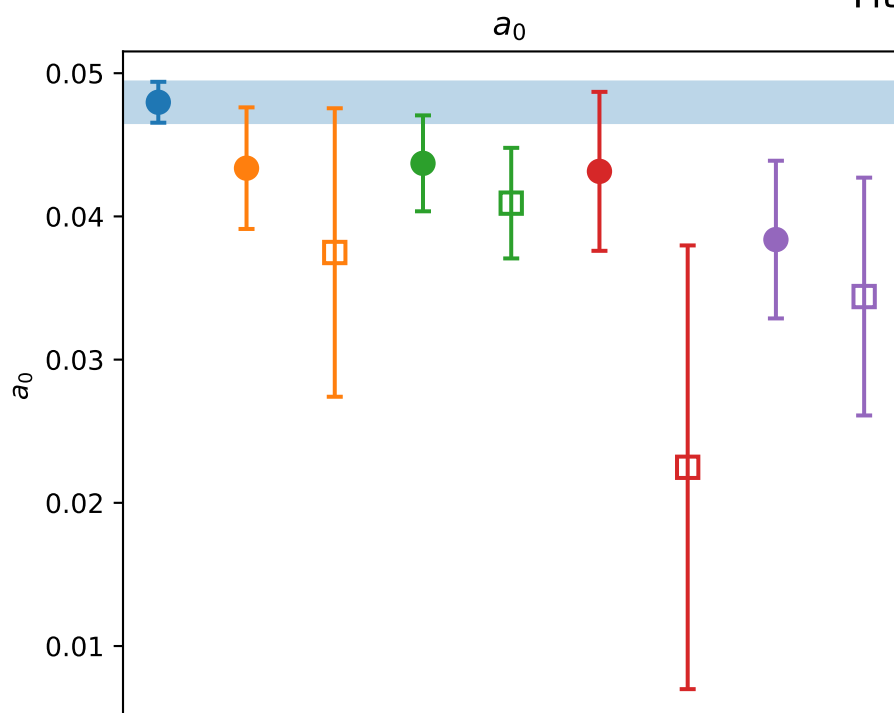
MLP: RM



TRANSFORMER: RM



Two_pt_ML_OTSP_D_gold_to_nongold_Experiment
Fit Parameters



Two_pt_ML_OTSP_D_gold_to_nongold_Experiment

Fit Results

Bias Corrected

Model	a_0	E_0	χ^2/dof	Q
Truth	0.04797(0.00143)	0.84123(0.00433)	0.80	0.755
CNN	0.04337(0.00425)	0.83153(0.00867)	0.52	0.954
GBR	0.04371(0.00335)	0.83290(0.00725)	1.23	0.215
MLP	0.04315(0.00555)	0.83147(0.01039)	1.46	0.099
TRANSFORMER	0.03838(0.00551)	0.82157(0.01151)	0.84	0.648

Ratio Method

Model	a_0	E_0	χ^2/dof	Q
Truth	0.04797(0.00143)	0.84123(0.00433)	0.80	0.755
CNN+RM	0.03748(0.01007)	0.81716(0.02215)	0.96	0.497
GBR+RM	0.04093(0.00386)	0.82679(0.00891)	1.34	0.135
MLP+RM	0.02248(0.01549)	0.77983(0.04563)	0.94	0.501
TRANSFORMER+RM	0.03440(0.00830)	0.80967(0.01952)	0.88	0.601

Fit Results Summary

Model	Method	dE0	dE1	a0	a1	χ^2/dof	Q
Truth	-	0.8412(43)	0.1999(554)	0.0480(14)	0.0033(40)	0.802	0.755
CNN	BIAS	0.8315(87)	0.2000(416)	0.0434(42)	0.0034(30)	0.524	0.954
CNN	RATIO	0.8172(221)	0.2000(832)	0.0375(101)	0.0049(88)	0.963	0.497
GBR	BIAS	0.8329(73)	0.1999(554)	0.0437(33)	0.0033(40)	1.227	0.215
GBR	RATIO	0.8268(89)	0.1998(554)	0.0409(39)	0.0063(76)	1.343	0.135
MLP	BIAS	0.8315(104)	0.1999(1386)	0.0431(56)	0.0033(100)	1.461	0.099
MLP	RATIO	0.7798(456)	0.2000(832)	0.0225(155)	0.0060(107)	0.939	0.501
TRANSFORMER	BIAS	0.8216(115)	0.2000(693)	0.0384(55)	0.0033(50)	0.840	0.648
TRANSFORMER	RATIO	0.8097(195)	0.2000(693)	0.0344(83)	0.0062(94)	0.878	0.601